QUANTUM SCIENCE AND TECHNOLOGY

UNIT 2

Quantum Information Theory

Study Material

Course Outcome (CO2):
Analyze quantum phenomena like superposition and entanglement

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1. Introduction to Quantum Information Theory

Quantum information theory studies how information is represented, processed, transmitted and measured using the laws of quantum mechanics. Unlike classical information, which is built from bits that are always definitely 0 or 1, quantum information is built from qubits — quantum systems that can exist in superpositions of 0 and 1, and that can be entangled with one another. This unit builds the mathematical and conceptual foundation of quantum information: qubit representation, superposition, entanglement, Dirac notation, tensor products, Bell states, quantum gates, circuit notation, measurement, teleportation, the no-cloning theorem, and an introduction to quantum state tomography.

1.1 Classical vs. Quantum Information

The essential differences between classical and quantum information are summarised below.

Classical Information	Quantum Information
Uses classical bits (0 or 1).	Uses qubits, which can be 0, 1, or a superposition of both.
A bit exists in only one state at a time.	A qubit can exist in multiple states simultaneously due to superposition.
Bits are independent of one another.	Qubits can become entangled and remain correlated even when separated.
Processed using classical logic gates (AND, OR, NOT).	Processed using quantum gates (Hadamard, Pauli-X, CNOT, etc.).
Copying data is easy and exact.	An unknown quantum state cannot be copied exactly (No-Cloning Theorem).
Used in conventional computers and digital devices.	Used in quantum computers for solving certain complex problems more efficiently.

2. Qubit Representation Using the Bloch Sphere

A qubit is the fundamental unit of quantum information. While a classical bit is restricted to the value 0 or 1, a qubit's general state is a linear combination (superposition) of the two basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle\,,\ \text{where } \alpha, \beta \in \mathbb{C} \text{ and } |\alpha|^2 + |\beta|^2 = 1$$

Here α and β are complex probability amplitudes; $|\alpha|^2$ is the probability of measuring $|0\rangle$ and $|\beta|^2$ is the probability of measuring $|1\rangle$. Because probabilities must sum to one, the amplitudes obey the normalisation condition above.

2.1 The Bloch Sphere

The Bloch sphere, named after physicist Felix Bloch, is a geometric representation of the pure-state space of a single qubit. It is a unit 2-sphere in which:

- Every point on the surface corresponds to a pure quantum state; interior points correspond to mixed states.
- The North Pole represents the classical state $|0\rangle$ and the South Pole represents $|1\rangle$.
- Longitude (the angle ϕ) represents the relative phase between α and β.
- Latitude (the angle θ) represents the relative amplitude between α and β.

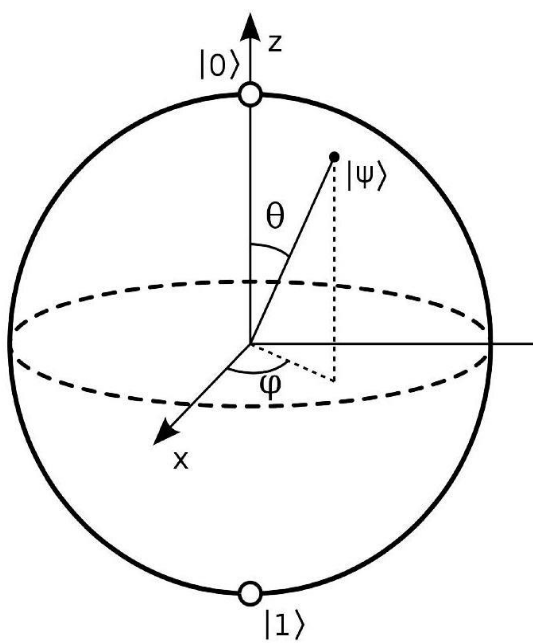


Figure 2.1: The Bloch sphere — a qubit state $|\psi\rangle$ is a unit vector specified by angles ϑ and φ.

Using spherical coordinates (θ, ϕ), any pure single-qubit state can be written, up to a global phase, as:

$$|\psi\rangle = \cos(\vartheta/2)\,|0\rangle + e^{i\varphi}\sin(\vartheta/2)\,|1\rangle$$

With this parametrisation, the poles correspond to the classical basis states:

$$|0\rangle = (\cos(0/2), \sin(0/2)) = (1, 0) \qquad |1\rangle = (\cos(\pi/2), \sin(\pi/2)) = (0, 1)$$

Because $|0\rangle$ and $|1\rangle$ form an orthonormal basis, any other qubit state can be expressed as a linear combination of these two classical states — this is precisely the principle of superposition, discussed next.

2.2 Simplified (Real-Field) View

If the global phase is ignored and ϕ is fixed, the qubit state can be reduced to a single angle θ measured from the $|0\rangle$ axis, simplifying the sphere to a circle while preserving the physically measurable amplitude information:

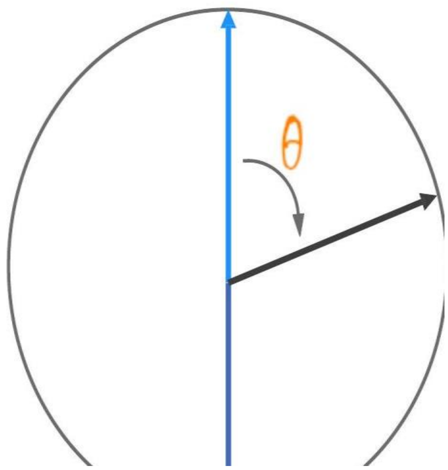


Figure 2.2: Simplified real-field (circle) view of the Bloch sphere, parametrised by ϑ alone.

In this reduced picture, $|\psi\rangle = (\alpha, \beta) = (\cos(\theta/2), \sin(\theta/2))$, with $|0\rangle$ at $\theta = 0$ and $|1\rangle$ at $\theta = \pi$. This view is useful for building intuition about single-qubit rotations before introducing the full complex phase ϕ.

3. Quantum Superposition

Superposition is the fundamental principle of quantum mechanics stating that a qubit can exist in a combination of the $|0\rangle$ and $|1\rangle$ states simultaneously, until it is measured. The general superposition state is written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

- α and β are complex probability amplitudes.
- $|\alpha|^2$ and $|\beta|^2$ give the probabilities of measuring the qubit in state $|0\rangle$ or $|1\rangle$ respectively.
- A classical bit can only be 0 or 1 — never both.
- A qubit in superposition is effectively "both 0 and 1 at once", which is what allows quantum computers to process many possibilities in parallel.

Important: even though a qubit can be mathematically written as a superposition, it is never possible to directly observe the superposition itself — measurement always yields a single classical outcome (0 or 1), and the act of measuring destroys the superposition. This is known as collapse of the wave function.

4. Quantum Entanglement

Entanglement occurs when two or more qubits become correlated in such a way that the state of one qubit cannot be described independently of the state of the other(s) — even when the qubits are separated by large distances. Measuring one entangled qubit instantaneously affects the outcome of measuring its partner.

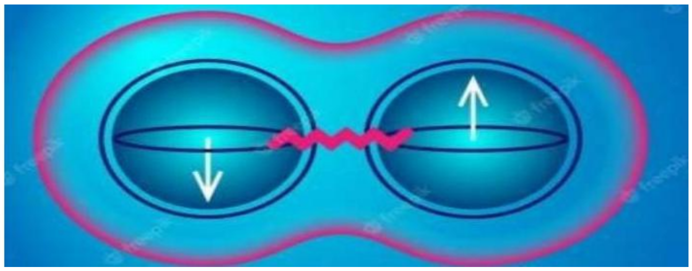


Figure 4.1: Two entangled particles behave as a single connected system.

A simple illustration uses electron spin. Spin is an intrinsic property of a particle, similar to a tiny compass needle pointing in some direction. If two electrons are entangled such that their total spin must add up to zero, then:

- If Particle A is measured as "spin-up" along an axis, Particle B must be "spin-down" along the same axis.
- This correlation holds no matter how far apart the particles are — a property Einstein famously called "spooky action at a distance" (see the EPR paradox, Section 9).

5. Measurement of a Qubit

When a qubit is measured, its superposition collapses into one of its two possible basis states (0 or 1), with a probability determined by the squared magnitude of the corresponding amplitude. Measurement (observation) projects the quantum state onto one of the basis vectors and destroys the superposition, yielding a classical outcome.

$$P(0) = |\alpha|^2 \qquad P(1) = |\beta|^2$$

This measurement postulate is central to how classical information is extracted from a quantum system, and is revisited formally in Section 15 (Measurement in the Computational Basis).

5.1 Worked Example

Consider a qubit prepared in the state $|\psi\rangle = 0.866|0\rangle + 0.5|1\rangle$ (i.e. $\alpha \approx 0.866$, $\beta = 0.5$). The measurement probabilities are:

$$P(0) = (0.866)^2 \approx 0.75 = 75\% \qquad P(1) = (0.5)^2 = 0.25 = 25\%$$

Note that $P(0) + P(1) = 1$, confirming the state is correctly normalised. If this qubit is measured, the outcome will be 0 about 75% of the time and 1 about 25% of the time across many identical trials — but any single measurement yields only one definite classical bit, and the original superposition is destroyed.

6. Dirac Notation (Bra–Ket)

Dirac notation, introduced by physicist Paul Dirac, is a compact notation used to describe quantum states and operations using linear algebra.

6.1 The Ket $|\psi\rangle$

A ket represents a state vector in a Hilbert space and is written as a column vector. For example:

$$|0\rangle = (1, 0)^T \qquad |1\rangle = (0, 1)^T \qquad |\psi\rangle = \alpha|0\rangle + \beta|1\rangle = (\alpha, \beta)^T, \quad |\alpha|^2+|\beta|^2=1$$

Analogy: a ket is like a seed placed into soil — it represents the starting state of the system.

6.2 The Bra $\langle\psi|$

A bra is the Hermitian conjugate of a ket: the complex-conjugate row vector. If $|\psi\rangle = (\alpha, \beta)^T$ then:

$$\langle\psi| = (\alpha^*, \beta^*)$$

Analogy: a bra is like the soil that receives the seed — it gives meaning to a ket when the two are combined.

6.3 Inner Product

The inner product combines a bra $\langle\phi|$ and a ket $|\psi\rangle$ to produce a single complex number $\langle\phi|\psi\rangle$, called the probability amplitude. It measures how similar (or how orthogonal) two quantum states are.

$$\langle 0|0\rangle = 1 \quad \text{(identical states} \rightarrow \text{result 1)} \qquad \langle 0|1\rangle = 0 \quad \text{(orthogonal states} \rightarrow \text{result 0)}$$

6.4 Outer Product

The outer product combines a ket $|\psi\rangle$ and a bra $\langle\phi|$ to form a matrix (operator) $|\psi\rangle\langle\phi|$. Outer products are used to build projection operators and quantum gates. For example, multiplying the column vector $|0\rangle$ by the row vector $\langle 0|$ gives:

$$|0\rangle\langle 0| = [[1,0],[0,0]]$$

7. Tensor Products and Composite Systems

A single qubit represents one quantum system. Quantum computers, however, perform computation using many qubits together. The tensor product ($\otimes$) is the mathematical operation used to combine two or more individual qubit states into one larger, composite quantum state. Unlike classical bits — which are simply listed side by side — quantum states must be combined using the tensor product.

7.1 Composite Systems

- One qubit → single quantum system → $|0\rangle$ or $|1\rangle$ (2 basis states).
- Two qubits → composite system → $|0\rangle \otimes |1\rangle = |01\rangle$ ($2^2 = 4$ basis states: $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$).
- Three qubits → composite system → $|1\rangle \otimes |0\rangle \otimes |1\rangle = |101\rangle$ ($2^3 = 8$ basis states).

In general, an n-qubit composite system lives in a Hilbert space of dimension 2^n.

7.2 Why Tensor Products Are Needed

Given two independent qubits $|\psi\rangle$ and $|\phi\rangle$, we cannot represent their combined state by simple addition. Instead we combine them using the tensor product operator $\otimes$

$$|\Psi\rangle = |\psi\rangle \otimes |\varphi\rangle$$

7.3 Worked Example — Composite State

Suppose the first qubit is in superposition and the second qubit is in the basis state $|0\rangle$:

7. Composite State Example

Suppose the first qubit is in superposition

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Second qubit is

$$|0\rangle$$

The composite state becomes

$$|\Psi\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes |0\rangle$$

Expanding,

$$|\Psi\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

This represents a two-qubit system in a superposition of two basis states.

Figure 7.1: Expanding the tensor product of a superposed qubit with $|0\rangle$.

The resulting two-qubit state is a superposition of two basis states, $|00\rangle$ and $|10\rangle$, each with equal probability amplitude $1/\sqrt{2}$.

7.4 Separable vs. Entangled States

A composite-system state is called separable if it can be written as the tensor product of individual qubit states, for example $(|0\rangle+|1\rangle)/\sqrt{2} \otimes |0\rangle$. Some composite states, however, cannot be factored into a tensor product of individual qubit states — for example:

$$|\Psi\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$$

Such states are called entangled states (see Section 8). In an entangled state, measuring one qubit immediately determines the outcome for the other, even though neither qubit has a well-defined individual state on its own.

8. Bell States

A Bell state is a special quantum state of two qubits in which the qubits are maximally entangled — measuring one qubit immediately determines the correlated outcome of the other, no matter how far apart they are. A simple definition: a Bell state is a pair of entangled qubits that always produce correlated measurement results.

The Four Bell States

The four Bell states are:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$
$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$
$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$
$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

Figure 8.1: The four Bell states, which together form the Bell basis.

8.1 Worked Example (Alice and Bob)

Suppose Alice and Bob each receive one qubit from the same Bell pair, then travel far apart. When Alice measures her qubit:

- If she measures 0, Bob's qubit is also found to be 0 when he measures his.
- If she measures 1, Bob's qubit is also found to be 1.

Even though they are far apart, their results are always correlated because the two qubits are entangled — entangled qubits behave as a single connected system.

9. EPR Paradox

The EPR paradox was proposed in 1935 by Albert Einstein, Boris Podolsky, and Nathan Rosen. It questions whether quantum mechanics gives a complete description of physical reality. In simple terms: two entangled particles remain correlated even when very far apart, and a measurement on one particle appears to instantly determine the state of the other.

9.1 Why Einstein Was Troubled

Einstein reasoned that no information or influence should be able to travel faster than the speed of light, so he called the apparent instantaneous correlation "spooky action at a distance". He argued that if quantum mechanics permitted this without contradiction, either information was travelling faster than light, or quantum mechanics was an incomplete theory (i.e. some "hidden variables" must pre-determine the outcomes).

9.2 Resolution — Bell's Theorem

Physicist John Bell later proposed Bell's Theorem, and subsequent experiments demonstrated that:

- Quantum entanglement is real — the particles are genuinely, strongly correlated.
- No information is actually transmitted faster than light — the individual measurement outcomes are random; the correlation can only be verified by later comparing results using ordinary (classical-speed) communication.

9.3 Key Points (Exam Perspective)

- EPR stands for Einstein, Podolsky and Rosen; proposed in 1935.
- It questions the completeness of quantum mechanics.
- Entangled particles remain correlated even when far apart.
- Einstein termed this "spooky action at a distance."
- Bell's Theorem and later experiments confirmed entanglement is real, while confirming that no faster-than-light signalling occurs.

10. Quantum Gates

Quantum gates are the building blocks of quantum circuits — basic operations applied to a small number of qubits. Unlike many classical logic gates, quantum gates are reversible and are mathematically represented by unitary matrices (a matrix U is unitary if U U† = I). Common single-qubit gates include the Hadamard gate, the three Pauli gates, phase-shift gates (S, T, and the general Rφ family), and general single-qubit rotations (U gates). Two-qubit controlled gates (CNOT, CY, CZ) and multi-qubit gates (SWAP, Toffoli) are used to create entanglement and implement more complex operations.

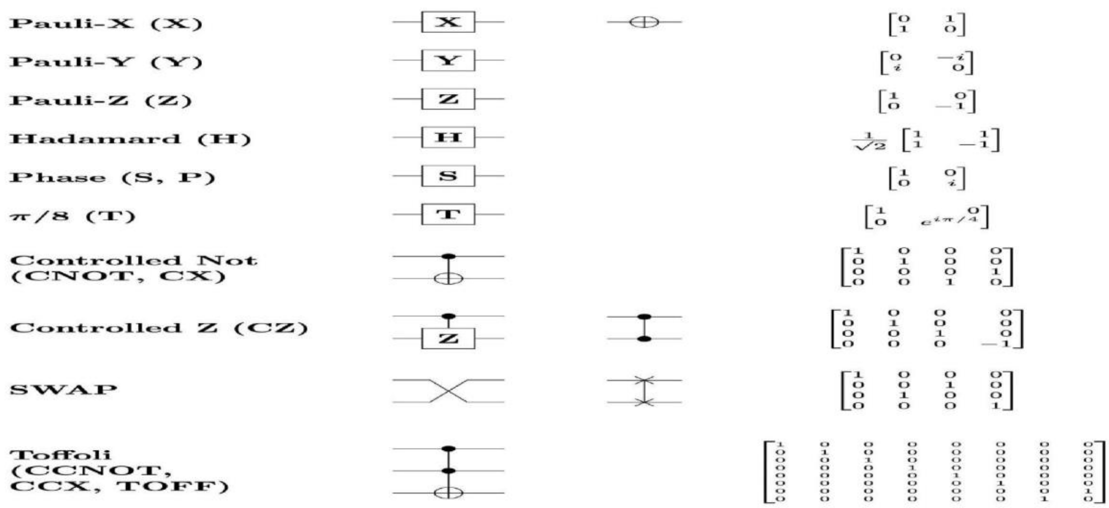


Figure 10.1: Common quantum gates — circuit symbol and matrix representation.

11. Pauli Gates: X, Y, Z

The three Pauli gates are the most fundamental single-qubit gates. Each corresponds to a 180° (π radian) rotation of the qubit's Bloch-sphere vector about one axis.

11.1 Pauli-X Gate

The X gate is the quantum analogue of the classical NOT gate: it flips $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$. On the Bloch sphere this corresponds to a rotation of π radians (180°) about the x-axis.

$$-\boxed{X}- = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = |1\rangle\langle 0| + |0\rangle\langle 1|$$

Figure 11.1: The Pauli-X gate — matrix form and outer-product form.

11.2 Pauli-Y Gate

The Y gate represents a rotation of π radians about the y-axis. It transforms $|0\rangle \rightarrow i|1\rangle$ and $|1\rangle \rightarrow -i|0\rangle$.

$$Y = [[0, -i], [i, 0]] = i|1\rangle\langle 0| - i|0\rangle\langle 1|$$

11.3 Pauli-Z Gate

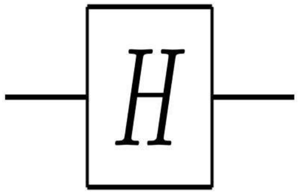
The Z gate is a special case of the phase-shift gate with $\phi = \pi$ (180°). It represents a rotation of π radians about the z-axis. It leaves $|0\rangle$ unchanged but transforms $|1\rangle$ to $-|1\rangle$.

$$Z = [[1, 0], [0, -1]] \qquad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle$$

12. Hadamard, Phase, and T Gates

12.1 Hadamard (H) Gate

The Hadamard gate is one of the most important gates in quantum computing because it creates superposition from a classical basis state. Geometrically, it can be understood as a π rotation about the x-axis followed by a $\pi/2$ (clockwise) rotation about the y-axis.



$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Figure 12.1: The Hadamard gate — circuit symbol and matrix.

Applying H to the basis states gives equal-superposition states:

$$H|0\rangle = (|0\rangle+|1\rangle)/\sqrt{2} = |+\rangle \qquad H|1\rangle = (|0\rangle-|1\rangle)/\sqrt{2} = |-\rangle$$

12.2 Phase-Shift Gate (Rφ)

This is a family of single-qubit gates that leave the basis state $|0\rangle$ unchanged and map $|1\rangle$ to $e^{i\phi}|1\rangle$. The probability of measuring $|0\rangle$ or $|1\rangle$ is unaffected, but the relative phase of the state is modified. Geometrically this corresponds to tracing a line of latitude on the Bloch sphere by an angle ϕ.

$$R\phi = [[1, 0], [0, e^{i\varphi}]]$$

12.3 S and T Gates

The S gate (also called the phase gate, P) is the special case $\phi = \pi/2$. The T gate is the special case $\phi = \pi/4$, and is sometimes referred to as the $(Z)^{1/4}$ gate. The conjugate transpose of T is written $T^\dagger$.

$$S = [[1, 0], [0, i]] \qquad T = [[1, 0], [0, e^{i\pi/4}]]$$

13. Controlled Gates

Controlled gates act on two (or more) qubits: at least one control qubit and one target qubit. The gate operates on the target qubit only when the control qubit is in a specific state (usually $|1\rangle$); otherwise the target is left unchanged.

13.1 CNOT (Controlled-X, CX) Gate

The CNOT gate leaves the control qubit unchanged, and applies a Pauli-X (bit-flip) to the target qubit whenever the control qubit is $|1\rangle$; it leaves the target unchanged when the control is $|0\rangle$.

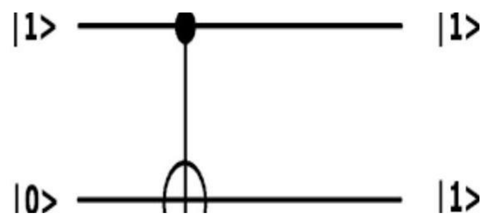


Figure 13.1: CNOT circuit — control $|1\rangle$ flips the target from $|0\rangle$ to $|1\rangle$.

$$CX\begin{bmatrix}1\\0\\1\\0\end{bmatrix}=\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&0&1\\0&0&1&0\end{bmatrix}\begin{bmatrix}1\\0\\1\\0\end{bmatrix}=\begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

Figure 13.2: CNOT acting on the state $|10\rangle$, producing $|11\rangle$ (matrix computation).

Because the CNOT gate acts on two qubits, its matrix representation is a 4×4 unitary matrix:

CNOT = [[1,0,0,0], [0,1,0,0], [0,0,0,1], [0,0,1,0]]

13.2 Controlled-Y (CY) and Controlled-Z (CZ) Gates

Just as CNOT applies an X to the target whenever the control is $|1\rangle$, the controlled-Y (CY) gate applies a Y in the same circumstance, and the controlled-Z (CZ) gate applies a Z. CZ is symmetric in its two qubits — it is often drawn as two control dots connected by a line, since it makes no difference which qubit is called "control" and which is "target".

14. Quantum Circuit Models and Notation

A quantum circuit is a visual representation of a quantum algorithm, analogous to a schematic for a quantum program. A quantum circuit consists of qubits (drawn as horizontal wires) and quantum gates applied to those qubits. Time flows from left to right: qubits begin on the left in some initial state; gates are applied left to right; measurements are typically shown on the far right (though they may also appear in the middle of a circuit).

14.1 Reading a Circuit — Bell-State Example

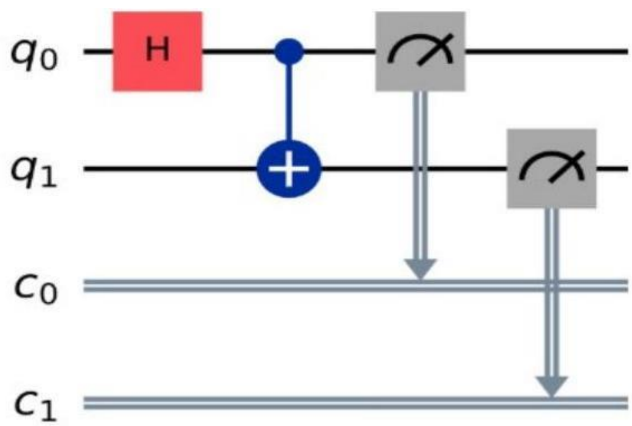


Figure 14.1: A circuit that creates and measures a Bell pair.

- q_0 and q_1 are the two qubits used in the circuit, both initialised to $|0\rangle$.
- The H (Hadamard) gate puts q_0 into an equal superposition of $|0\rangle$ and $|1\rangle$.
- The CNOT gate uses q_0 as control and q_1 as target, entangling the two qubits into a Bell state.
- The measurement symbol (M) measures both qubits, producing classical results 0 or 1.
- c_0 and c_1 are classical registers that store the measurement results of q_0 and q_1 respectively.

Common circuit symbols: a labelled box (e.g. X, H, Z) represents a single-qubit gate; a solid dot connected by a vertical line to a $\oplus$symbol represents a CNOT; a grey box with a dial/arrow icon represents a measurement, whose result is routed to a double-line classical wire.

14.2 Single-Gate Circuit Example — Identity Gate

The identity gate I is the simplest possible quantum gate: it leaves the qubit's state completely unchanged.

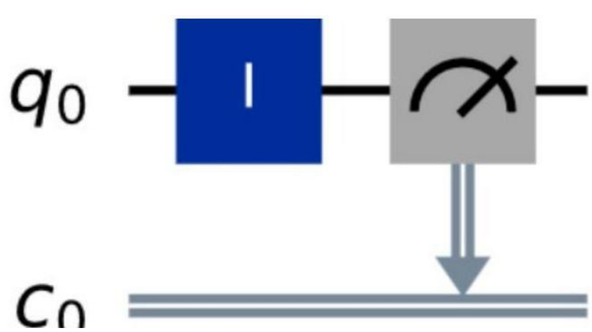


Figure 14.2: Applying the identity gate to q_0, then measuring into classical register c_0.

$$I|0\rangle = [[1,0],[0,1]]\cdot[1,0]^T = [1,0]^T = |0\rangle \qquad I|1\rangle = [[1,0],[0,1]]\cdot[0,1]^T = [0,1]^T = |1\rangle$$

The blue box labelled "I" symbolises the identity gate; the grey box with the dial icon indicates a measurement of qubit q_0, whose classical result (0 or 1) is stored in the classical register c_0 (drawn as a double line).

15. Measurement in the Computational Basis

The computational basis for a single qubit is the set $\{|0\rangle, |1\rangle\}$. When a qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ is measured in this basis, the measurement probabilities are fixed by the amplitudes:

$$P(0) = |\alpha|^2 \qquad P(1) = |\beta|^2 \qquad (\text{normalisation: } |\alpha|^2 + |\beta|^2 = 1)$$

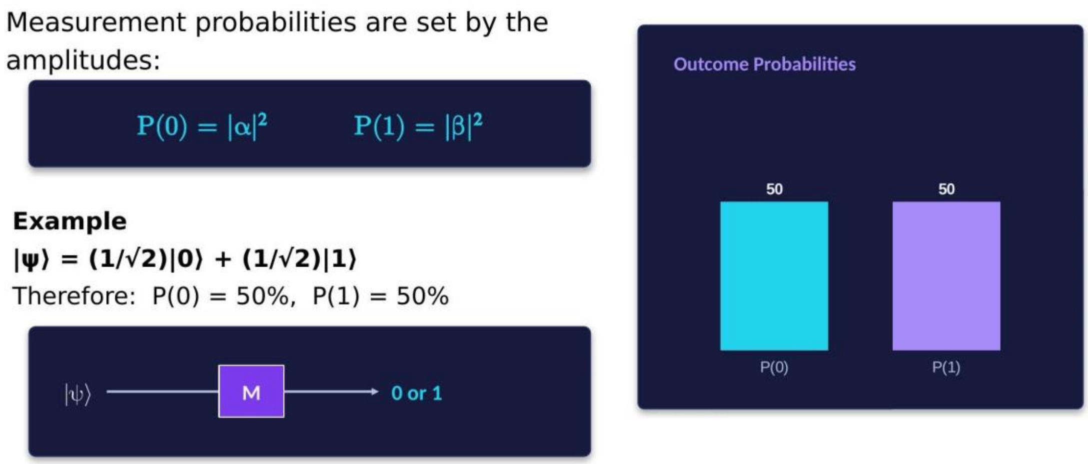


Figure 15.1: Measurement of an equal-superposition state gives a 50/50 outcome distribution.

Worked example: for $|\psi\rangle = (1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$, we get P(0) = 50% and P(1) = 50%, as shown in the outcome-probability chart above.

16. Quantum Teleportation

Quantum teleportation transfers an unknown quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ from a sender (Alice) to a receiver (Bob), using a pre-shared entangled qubit pair and a classical communication channel. Teleportation does not transport matter or a physical object — only quantum information is transferred, and it requires two classical bits of communication.

16.1 Step-by-Step Circuit

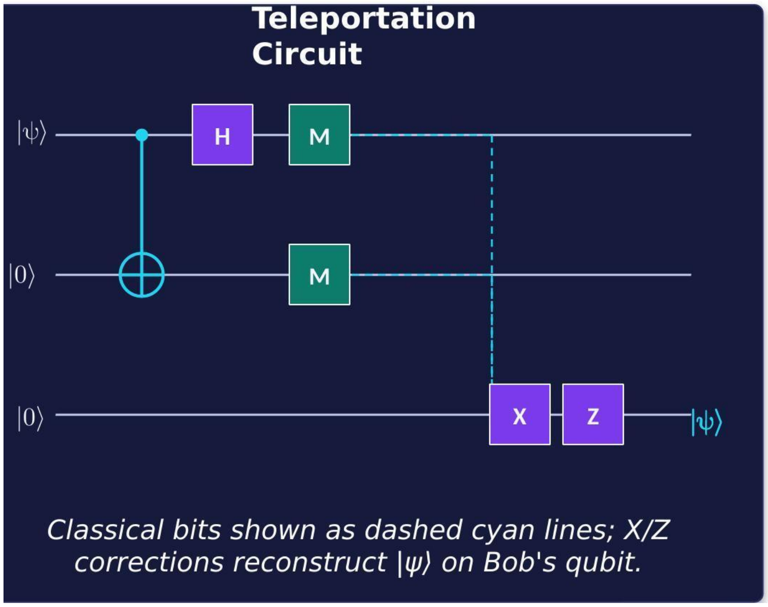


Figure 16.1: The quantum teleportation circuit.

- Alice holds the unknown state $|\psi\rangle$ that she wants to send to Bob.
- Alice and Bob pre-share an entangled Bell pair (one qubit each).
- Alice applies a CNOT (using $|\psi\rangle$ as control and her half of the Bell pair as target), then a Hadamard gate to $|\psi\rangle$.
- Alice measures both of her qubits, obtaining two classical bits.
- She sends these 2 classical bits to Bob over an ordinary classical channel.
- Depending on the bits received, Bob applies an X and/or Z correction to his qubit, reconstructing the exact original state $|\psi\rangle$.

Teleportation relies fundamentally on entanglement plus classical communication; no information travels faster than light, since Bob must wait to receive Alice's classical bits before he can complete the reconstruction.

17. The No-Cloning Theorem

The no-cloning theorem states that an arbitrary, unknown quantum state cannot be perfectly copied. While classical data can always be copied exactly, no quantum operation can take an unknown state $|\psi\rangle$ and produce two independent copies of it.

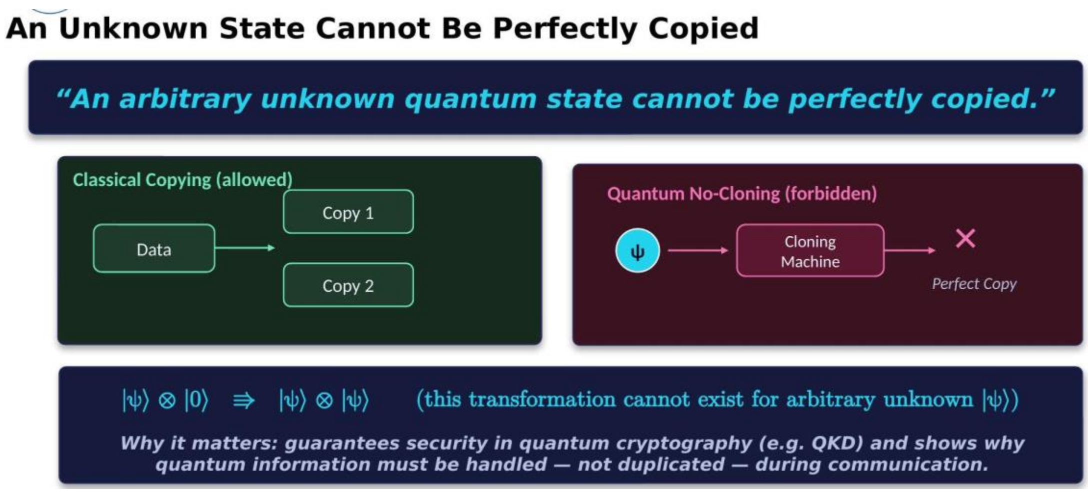


Figure 17.1: Classical copying is allowed; quantum cloning of an unknown state is forbidden.

$$|\psi\rangle \otimes |0\rangle \nRightarrow |\psi\rangle \otimes |\psi\rangle \quad \text{(no unitary transformation performs this for arbitrary unknown } |\psi\rangle\text{)}$$

Why it matters: the no-cloning theorem guarantees security in quantum cryptography protocols such as Quantum Key Distribution (QKD), since an eavesdropper cannot copy a transmitted qubit without disturbing it. It also explains why quantum information must be handled and transferred (as in teleportation) rather than duplicated during communication.

17.1 Connecting Teleportation and No-Cloning

Quantum teleportation is fully consistent with the no-cloning theorem:

- Teleportation never creates a second copy of the state — Alice's original qubit is destroyed the moment she measures it.
- Bob's qubit only becomes $|\psi\rangle$ after he receives Alice's classical bits and applies the appropriate correction.
- At every instant, exactly one copy of $|\psi\rangle$ exists in the combined system — so no violation of no-cloning occurs.

Teleportation	Cloning
Only one, existing at the end	Two identical copies
Destroyed on measurement	Preserved
Yes	No (impossible for unknown states)

18. Quantum State Tomography (Introductory)

We cannot directly "see" a complete quantum state — a single measurement only reveals one classical bit of information and irreversibly disturbs the state. Quantum state tomography is the technique used to reconstruct/estimate an unknown quantum state indirectly, using statistics gathered from many repeated measurements on many identically-prepared copies of the state.

18.1 Tomography Pipeline

- Prepare many identical copies of the unknown state.
- Perform repeated measurements on these copies.
- Collect statistical outcomes across all the measurements.
- Use the statistics to reconstruct (estimate) the quantum state.

Medical analogy: just as a CT scan reconstructs an internal image from many external measurements taken from different angles, quantum state tomography reconstructs a hidden quantum state from many measurement statistics.

18.2 Measuring in Different Bases

A single qubit's state cannot be fully revealed by measurements in only one basis. Measuring along the X, Y and Z axes separately gives expectation values that together locate the state uniquely on the Bloch sphere — this is the essence of single-qubit tomography.

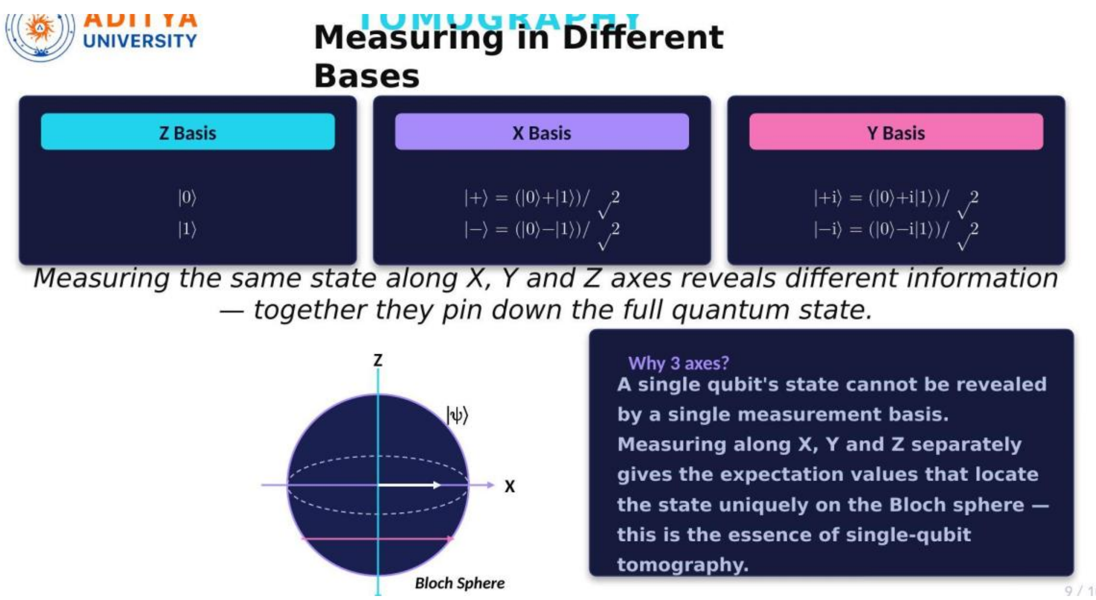


Figure 18.1: Measuring a qubit along the Z, X and Y bases reveals complementary information.

Z basis: $\{|0\rangle, |1\rangle\}$ *X basis:* $\{|+\rangle=(|0\rangle+|1\rangle)/\sqrt{2}, |-\rangle=(|0\rangle-|1\rangle)/\sqrt{2}\}$ *Y basis:* $\{|+i\rangle=(|0\rangle+i|1\rangle)/\sqrt{2}, |-i\rangle=(|0\rangle-i|1\rangle)/\sqrt{2}\}$

19. Summary

This unit developed the foundations of quantum information theory, moving from single-qubit representation to multi-qubit systems and practical quantum-information protocols:

- Classical bits vs. qubits, and the Bloch-sphere geometric picture of a qubit state.
- Superposition and entanglement as the two properties that fundamentally distinguish quantum from classical information.
- Dirac (bra-ket) notation, inner and outer products, and how tensor products build composite multi-qubit systems.
- Bell states and the EPR paradox, and how Bell's theorem resolved Einstein's "spooky action at a distance" concern.
- Quantum gates (Pauli-X/Y/Z, Hadamard, Phase, T, and controlled gates) and how they are combined into quantum circuits.
- Measurement in the computational basis, quantum teleportation, and the no-cloning theorem.
- An introduction to quantum state tomography for reconstructing an unknown quantum state.

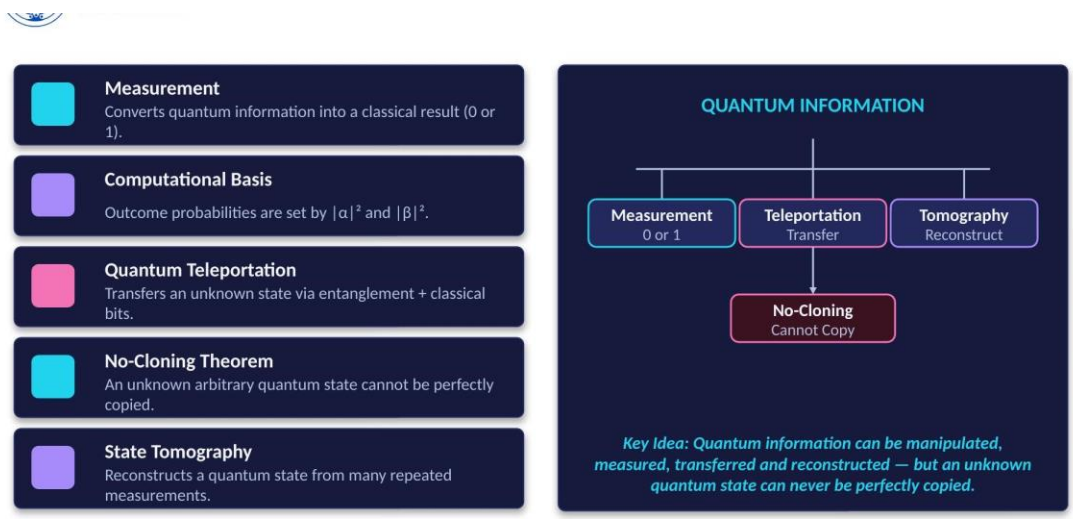


Figure 19.1: Key ideas of Unit 2 at a glance.

Key takeaway: quantum information can be manipulated, measured, transferred, and reconstructed — but an unknown quantum state can never be perfectly copied.